e^+e^-, that is given by [26]

$$\frac{dN_e}{dE_e} = N_e C_0 \left(\frac{4E_e^2 - m_X^2}{E_0^2}\right)^{\beta/2} e^{-(1+\beta)\frac{2E_e}{E_0}} ,$$
$$C_0 = \frac{2\sqrt{\pi}\left(\frac{1+\beta}{2m_X}\right)^{\frac{1+\beta}{2}} E_0^{\frac{\beta-1}{2}}}{K_{\frac{1+\beta}{2}}\left((1+\beta)\frac{m_X}{E_0}\right)\Gamma\left(1+\frac{\beta}{2}\right)} , \tag{1}$$

where E_e is the emitted lepton energy and we assume that FIPs are emitted with a modified blackbody spectrum, where E_0 is related to the FIP average energy, its mass is $m_X > 2m_e$, β is the spectral index, $K_{\frac{1+\beta}{2}}$ is the modified Bessel function of the second kind of order $(1+\beta)/2$, Γ is the Euler-Gamma function and this flux is normalized such that

$$\int_{m_X/2}^{\infty} dE_e \frac{dN_e}{dE_e} = N_e . \tag{2}$$

Here, we use N_e to denote the number of electrons, which is equal to the number of positrons, produced in a SN explosion via FIP decays, i.e. $N_e = N_{e^+} = N_{e^-}$. The simple prescription in Eq. (1) does not depend on the type of FIP model. This flux is obtained by assuming a FIP decaying into an electron-positron pair. Thus, it cannot be strictly valid for sterile neutrinos, which has more involved decay channels. However, by changing the parameters in this simple prescription for the injected lepton flux enabled us to place constraints on the electron-coupling of different electrophilic FIPs in a model-independent way.

For the sake of completeness, in this work we go beyond this simplification and we derive the full positron injection spectra for the cases of ALPs, DPs and sterile neutrinos (coupling to ν_τ and ν_μ), evaluating SN FIP emission spectra for each specific case.

For definiteness, in the following all the FIP emission rates will be compute by employing the state-of-the-art 1D spherical-symmetric `GARCHING` group's SN model SFHo-s18.8 provided in Ref. [38], already used in Refs. [11, 39–45]. This model is developed from a stellar progenitor with mass 18.8 $M_\odot$ [46] and leads to a NS with baryonic mass 1.35 $M_\odot$. The simulation is based on the neutrino-hydrodynamics code `PROMETHEUS-VERTEX` [47], taking into account all neutrino reactions relevant for core-collapse SNe [48–51]. Moreover, it also accounts for a 1D treatment of PNS convection via a mixing-length description of the convective fluxes [2] as well as muon physics [50]. For the sake of clarity, we highlight that the choice of a given SN model in the analysis introduce unavoidable uncertainties on FIP fluxes, which are typically sensitive to the SN core temperatures and densities. However, we remark that the SN model adopted is characterized by the coldest PNS profile among those of the `GARCHING` group's archive [38] (see Ref. [45] e.g). Thus, since a high SN core temperature significantly power FIP emission, our results have to be considered as conservative.

Finally, we point out that, since we work in the setup where FIPs couple weakly with ordinary matter, we neglect possible feedback of FIP emission on the SN profile employed (see Refs. [52–55] for some works on possible impact of FIPs on the SN explosion mechanism.

ALPs coupled to electrons

Axion interactions with electrons are described by the following Lagrangian [1]

$$\mathcal{L}_{ae} = \frac{g_{ae}}{2\,m_e}\bar{\psi}_e\gamma^\mu\gamma^5\psi_e\,\partial_\mu a , \tag{3}$$

where ψ_e and a are, respectively, the electron and axion fields, m_e is the electron mass, and g_{ae} is the dimensionless axion-electron coupling. Starting from this Lagrangian, it is possible to realize that electron-ion bremsstrahlung is the dominant axion production process in core-collapse SNe for a vast range of ALP masses. The number of axions produced per unit volume and time can be evaluated as [56, 57]

$$\begin{aligned}\frac{d^2 n_a}{dt\,d\omega_a} =& 2\pi \int \frac{2d^3\mathbf{p}_i}{(2\pi)^3 2E_i}\frac{2d^3\mathbf{p}_f}{(2\pi)^3 2E_f}\frac{|\mathbf{p}_a|}{(2\pi)^3} \\ & (2\pi)\delta(E_i - E_f - \omega_a)\,|\mathcal{M}|^2 f_i(1-f_f) = \\ =& \frac{1}{64\pi^6}\int d\cos\theta_{ia}\,d\cos\theta_{if}\,d\delta\,dE_f \\ & |\mathbf{p}_i||\mathbf{p}_f||\mathbf{p}_a||\mathcal{M}|^2 f_i(1-f_f) ,\end{aligned} \tag{4}$$

where ω_a, E_i and E_f are the energies of the axion, initial and final electrons respectively; $f_{i,f}$ are the electron distribution functions; θ_{ia}, $\theta_{if} \in [0,\pi]$ are the angles between the initial electron and the axion and the final electron momenta respectively; $\delta \in [0, 2\pi]$ is the angle between the two planes determined by the vectors $\mathbf{p}_i - \mathbf{p}_a$ and $\mathbf{p}_i - \mathbf{p}_f$ and the matrix element averaged over the electron spins and summed over all the target ions is [56]

$$\begin{aligned}|\mathcal{M}|^2 = \frac{1}{4}\sum_j n_j \sum_{\rm s} |\mathcal{M}_j|^2 = \frac{g_{ae}^2 e^2}{2}\frac{k_S^2 T}{|\mathbf{q}|^2(|\mathbf{q}|^2 + k_S^2)} \\ \left[2\omega_a^2\frac{p_i\cdot p_f - m_e^2 - K\cdot p_a}{(p_i\cdot p_a)(p_f\cdot p_a)} + 2 - \frac{p_f\cdot p_a}{p_i\cdot p_a} - \frac{p_i\cdot p_a}{p_f\cdot p_a}\right] ,\end{aligned} \tag{5}$$

where $K = p_f - p_i$, for massless axions. The complete result for massive axions is shown in Appendix A of Ref. [56].

Nevertheless, electron-positron fusion $e^+ + e^- \to a$ could become dominant over bremsstrahlung in certain mass ranges. The electron fusion ALP emission spectrum per unit volume is given by [57]

$$\frac{d^2 n_a}{d\omega_a\,dt} = \frac{g_{ae}^2 m_a^2}{16\pi^3}\int_{E_{\rm min}}^{E_{\rm max}} dE_+ f_+ f_- , \tag{6}$$